Automatic Cricket Video Segmentation By Player Identity And Event Type

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***Abstract*—Cricket match video analysis is important for performance evaluation, coaching, and viewer engagement. However, locating player-specific actions and key match events in full-length cricket recordings usually requires manual tagging or lengthy video review. To address this limitation, this research proposes an automated cricket video segmentation system that indexes match videos by player identity and major match events. The proposed multimodal pipeline extracts scoreboard information using EasyOCR to obtain player names and match context. Key events such as FOUR, SIX, and WICKET are detected through computer-vision-based umpire signal recognition using EfficientNet-B0 for umpire detection and ResNet-18 for pose-based event classification. The extracted text is further processed using Named Entity Recognition (NER) to identify player entities. Detected events and player information are synchronized using timestamps to associate players with specific match moments. The processed outputs are integrated into a web-based platform that supports player-wise filtering, event-based navigation, and semantic search using MiniLM sentence embeddings with FAISS vector similarity retrieval. The proposed system generates structured, timestamped video segments, allowing users to directly access relevant match moments while reducing manual effort and improving retrieval efficiency.**

***Index Terms*—Cricket Video Analysis, Player Identification, Video Segmentation, EasyOCR, EfficientNet-B0, ResNet-18, Named Entity Recognition (NER), Semantic Search, FAISS, Sports Video Analytics.**

I. Introduction

Cricket is one of the most widely followed sports in the world. It produces a large volume of broadcast video content through international tournaments, domestic leagues, and televised matches. These videos contain valuable information for performance analysis, tactical decision-making, coaching, and viewer engagement. Traditionally, analysts and viewers depend on manual review or manually prepared highlights to locate specific player actions or key match events. However, reviewing full-length cricket matches is time-consuming and inefficient, especially when users need to find moments such as boundary shots, wickets, or player-specific performances [1], [5].

With the rapid growth of sports analytics and artificial intelligence, automated video understanding has become an important research area. Computer vision and natural language processing techniques have been widely applied to sports video analysis for tasks such as event detection, player recognition, and highlight generation [1], [2], [5], [6]. However, many existing approaches rely heavily on manually annotated datasets, which are costly and difficult to scale. In cricket broadcasts, visual cues such as scoreboard overlays and umpire signals provide useful contextual information that can be used to detect match events and identify player-related activities [7], [8].

In this research, an automated cricket video segmentation system is proposed to combine multiple modalities and generate structured event-based segments from match videos. The system extracts scoreboard information from video frames using EasyOCR [9], enabling the identification of player names and match context. Key cricket events such as FOUR, SIX, and WICKET are detected through umpire-signal recognition using EfficientNet-B0 for umpire detection [10] and ResNet-18 for pose-based event classification [11]. The extracted text is further processed using Named Entity Recognition (NER) techniques to identify player entities and associate them with detected match events [12].

To support efficient retrieval and exploration, detected events and player information are synchronized using timestamps and indexed for search. The processed outputs are integrated into a web-based platform that allows users to explore match videos through player-based filtering, event-based navigation, and semantic search. Semantic retrieval is implemented using MiniLM sentence embeddings with FAISS vector similarity search, enabling users to quickly locate relevant moments in full-length cricket videos [3], [4].

The main contributions of this research are summarized as follows:

1.Development of a multimodal cricket video segmentation pipeline combining OCR, computer vision, and natural language processing techniques.

2.Implementation of an umpire-signal-based event detection framework using EfficientNet-B0 and ResNet-18 for detecting key cricket events.

3.Integration of player identification using Named Entity Recognition and scoreboard OCR to associate players with detected match events.

4.Design of an interactive web-based platform that supports player-wise filtering, event navigation, and semantic search using MiniLM embeddings and FAISS.

The proposed system enables efficient indexing and retrieval of cricket match moments while reducing manual effort for analysts, coaches, and cricket fans.

II. Literature Review

Sports video analysis has attracted significant attention in computer vision and multimedia research because of its applications in performance analysis, event detection, and automated highlight generation. Early research mainly focused on extracting semantic information from sports videos using visual features and contextual cues. Wang and Parameswaran presented one of the early comprehensive studies in this area and discussed techniques for event detection, player tracking, and highlight extraction in sports broadcasts [5]. Their work highlighted the importance of semantic video understanding for efficient retrieval.

With the rise of deep learning, more advanced techniques have been introduced for detecting complex events and actions in sports videos. Modern approaches use convolutional neural networks and temporal models to identify meaningful events directly from broadcast footage. Recent studies show that deep-learning-based models improve the accuracy of action recognition and event detection compared with traditional feature-based methods [6]. These approaches support the automatic recognition of important game events such as goals, shots, and player actions across different sports.

Textual information embedded in broadcast videos has also been widely used to extract contextual information. Scoreboard overlays usually contain player names, scores, and match statistics. OCR techniques have therefore been applied to automatically extract such information from sports videos. Systems that combine OCR-based text extraction with computer vision can improve the accuracy of event detection and player identification by adding contextual support [7], [9].

In addition to visual and textual analysis, recent research has explored semantic search and embedding-based retrieval for multimedia indexing. Sentence embedding models such as Sentence-BERT generate meaningful vector representations of textual data, enabling similarity-based retrieval across large datasets [4]. High-performance vector search frameworks such as FAISS further support scalable similarity search for large embedding collections, making them suitable for multimedia retrieval systems [3].

Recent studies also emphasize the importance of multimodal learning in video analysis. By combining visual, textual, and contextual information, multimodal systems can improve the robustness and accuracy of event detection and content retrieval in sports videos [8]. Such methods allow systems to understand complex video content more effectively and produce richer semantic indexing for retrieval applications.

Despite these advances, many existing systems rely on large annotated datasets or focus on limited sports scenarios. In addition, relatively few approaches integrate event detection, player identification, and semantic video retrieval within a single interactive platform. Therefore, the proposed research introduces a multimodal framework that combines OCR-based scoreboard extraction, computer-vision-based umpire signal recognition, and semantic search to automatically segment cricket videos and retrieve player-specific match events.

III. Methodology

The proposed system automatically segments cricket match videos and retrieves player-specific events by combining computer vision, natural language processing, and semantic retrieval techniques. The overall framework processes broadcast cricket videos to detect key match events, identify players, and generate structured video segments that can be retrieved through a web-based platform. The main components of the methodology are video preprocessing, scoreboard text extraction, event detection, player identification, and semantic indexing [1], [8].

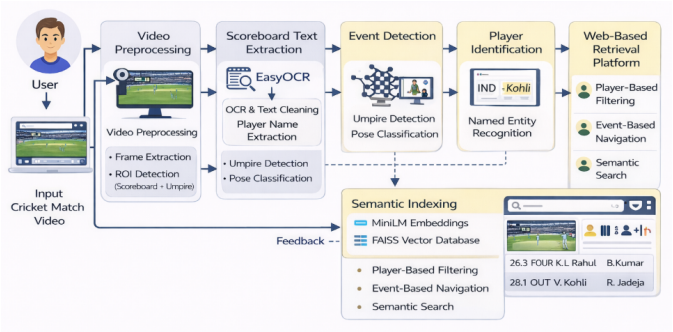

Fig. 1. High Level Architecture

A. Video Preprocessing

The input to the system is a full-length cricket match video obtained from broadcast recordings. The video is processed to extract frames at fixed time intervals. This reduces computational complexity while preserving the visual information needed for event detection and text recognition. Each frame is then analyzed to identify regions containing the scoreboard and the umpire area. This preprocessing stage helps the later modules focus only on relevant visual regions, improving both detection accuracy and efficiency [1], [6].

B. Scoreboard Text Extraction

Scoreboard overlays in cricket broadcasts contain useful contextual information such as player names, scores, and match statistics. In this work, scoreboard text is extracted from video frames using EasyOCR, a deep-learning-based optical character recognition framework [9]. The extracted

text is cleaned and processed to identify player names and relevant match context. Since OCR outputs may contain noise due to broadcast graphics, camera movement, and overlay variation, text normalization is applied to improve reliability. The extracted text is then used to estimate the batsmen and bowler present at a given moment in the match [7], [9].

C. Event Detection Using Umpire Signals

Key cricket events such as FOUR, SIX, and WICKET are detected using visual cues from umpire signals. The proposed event detection pipeline uses two deep learning models. First, an EfficientNet-B0 classifier is used to detect the presence of an umpire in a video frame [10]. Once an umpire is detected, a second model based on ResNet-18 classifies the specific pose corresponding to the event type [11]. This two-stage design improves detection accuracy by separating umpire detection from pose-based event classification. Detected events are recorded together with their timestamps to support accurate video segmentation [1], [6].

D. Player Identification

After extracting textual information and detecting events, the system associates players with the detected match events. Player identification is performed using Named Entity Recognition (NER) applied to the OCR-extracted text. The NER model identifies player entities from the text and maps them to the corresponding batsmen and bowler present in the frame. By combining OCR results with NER-based entity recognition, the system can determine which players are involved in specific match events more reliably [7], [12].

E. Timestamp Synchronization and Event Segmentation

Once events and player identities are detected, the system synchronizes them using timestamps. Each detected event is associated with the corresponding players and stored as a structured event record. These records include the event type, player information, and timestamp. Using this information, the original video is segmented into meaningful clips representing important moments such as boundaries and wickets [1], [8].

F. Semantic Indexing and Retrieval

To enable efficient retrieval of video segments, the extracted event data and textual descriptions are converted into vector representations using MiniLM sentence embeddings. These embeddings capture semantic relationships between textual queries and video segments. The embeddings are indexed using FAISS, a high-performance similarity search library that supports fast nearest-neighbor search in large embedding spaces [3], [4]. This allows users to perform semantic queries such as searching for specific players or events.

G. Web-Based Retrieval Platform

The final component of the system is a web-based platform that allows users to interact with indexed cricket videos. The platform provides functions such as player-based filtering, event-based navigation, and semantic search. Users can quickly retrieve relevant match segments by selecting players, events, or search queries. The platform displays timestamped video segments, allowing efficient navigation through full-length cricket matches [1], [8].

IV. Results And Discussion

This section presents the experimental results of the proposed cricket video segmentation system and discusses the effectiveness of its main components: umpire detection, umpire pose-based event classification, scoreboard text extraction, player identification, and semantic retrieval. The overall objective is to generate structured and searchable cricket video segments by combining visual event detection, OCR-based context extraction, and semantic indexing.

A. Dataset Summary

Three main datasets were used in the implementation of the proposed system: an umpire/non-umpire image dataset, an umpire pose dataset, and a labeled scoreboard OCR dataset. The umpire/non-umpire dataset was used to train the first-stage classifier that determines whether a frame contains an umpire. The pose dataset was used in the second stage to classify the detected umpire pose into one of the cricket event categories. In addition, the OCR dataset was used to support the extraction of textual match context from scoreboard regions.

The umpire/non-umpire dataset contained 1084 training images and 270 testing images. Within the training set, 446 images belonged to the umpire class and 638 images belonged to the non-umpire class. The testing set contained 111 umpire images and 159 non-umpire images. The umpire pose dataset contained 645 training images and 161 testing images. The training set included 202 out, 130 four, 162 no-action, and 151 six samples, while the testing set included 51 out, 32 four, 40 no-action, and 38 six samples.

The labeled scoreboard OCR dataset contained 3800 samples and was used to support the extraction of player-related textual information from broadcast scoreboard overlays.

These datasets allowed the system to model both the visual and textual characteristics required for event-based cricket video understanding. In particular, the use of scoreboard text is consistent with earlier work showing that OCR-based contextual extraction can improve semantic understanding in sports video analysis [7], [9], while the use of multimodal processing aligns with recent trends in robust video understanding systems [8].

TABLE I
Umpire Detection Dataset Distribution

Dataset Split	Umpire	Non-Umpire	Total
Training Set	446	638	1084
Test Set	111	159	270
Total	**557**	**797**	**1354**

B. Umpire Detection Performance

The first stage of the event detection pipeline is an umpire/non-umpire classifier based on EfficientNet-B0. This

TABLE II
UMPIRE POSE CLASSIFICATION DATASET DISTRIBUTION

Dataset Split	FOUR	SIX	OUT	No-Action	Total
Training Set	130	151	202	162	645
Test Set	32	38	51	40	161
Total	**162**	**189**	**253**	**202**	**806**

```
Classification Report:
              precision    recall  f1-score   support

  non_umpire       0.95      0.90      0.93       159
      umpire       0.87      0.94      0.90       111

    accuracy                           0.91       270
   macro avg       0.91      0.92      0.91       270
weighted avg       0.92      0.91      0.92       270
```

Fig. 2. Classification report of Umpire Detection Performance (EfficientNet-B0)

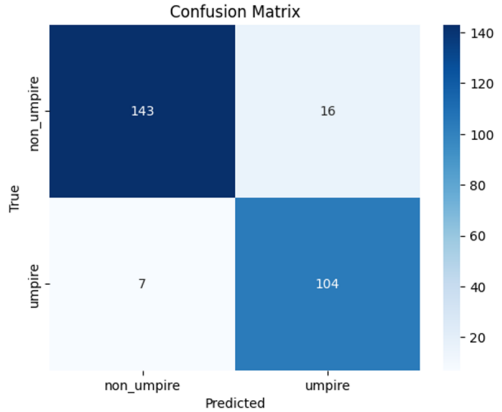

Fig. 3. Confusion matrix of Umpire Detection Performance (EfficientNet-B0)

model determines whether an extracted frame contains an umpire before pose-based classification is applied. The purpose of this stage is to reduce false event predictions by filtering irrelevant frames.

The classification report shown in Fig. 2 indicates that the model achieved an overall accuracy of 0.91 on the test set. For the non-umpire class, the precision, recall, and F1-score were 0.95, 0.90, and 0.93, respectively. For the umpire class, the model achieved a precision of 0.87, recall of 0.94, and F1-score of 0.90. The weighted average F1-score was 0.92.

The confusion matrix shown in Fig. 3 further shows that the model correctly classified 143 non-umpire samples and 104 umpire samples. Only 16 non-umpire images were misclassified as umpire, and 7 umpire images were misclassified as non-umpire. These results indicate that the first-stage detector is reliable for selecting candidate frames before pose classification.

This finding supports the effectiveness of the proposed two-stage event detection pipeline. Instead of classifying all broadcast frames directly into event categories, the system first isolates relevant umpire frames and then performs event classification. This design improves robustness and is consistent with deep-learning-based sports video analysis approaches that separate scene understanding into specialized subtasks [6], [10], [11].

C. Umpire Pose-Based Event Classification Performance

After detecting frames containing an umpire, the second stage of the proposed pipeline performs pose-based event classification. In this stage, a ResNet-18 model classifies umpire gestures into four event categories: FOUR, SIX, OUT, and NO-ACTION. This module identifies key cricket match events from umpire signals captured in broadcast frames.

The pose classification model was evaluated on a test dataset containing 161 samples. The model achieved an overall classification accuracy of 0.83. Fig. 4 presents the class-wise precision, recall, and F1-score values.

The FOUR class achieved a precision of 0.90, recall of 0.84, and F1-score of 0.87. The NO-ACTION class obtained a precision of 0.83, recall of 0.85, and F1-score of 0.84. The OUT class achieved a precision of 0.83, recall of 0.78, and F1-score of 0.81, while the SIX class obtained a precision of 0.79, recall of 0.87, and F1-score of 0.82.

Overall, the macro-average precision, recall, and F1-score were 0.84, while the weighted-average F1-score was 0.83. These results show that the model performs consistently across the different event classes.

The confusion matrix shown in Fig. 5 further illustrates the classification behavior of the model. It shows that 27 FOUR samples, 34 NO-ACTION samples, 40 OUT samples, and 33 SIX samples were correctly classified. Some misclassifications were observed between visually similar gestures, especially between OUT and SIX or FOUR and NO-ACTION, due to similarities in arm positioning and camera angle variations.

Despite these challenges, the overall results show that the ResNet-18 model successfully captures the visual patterns associated with umpire gestures. The use of pose-based event recognition is particularly effective for cricket because umpire signals provide a clear and structured representation of match events. Compared with general action recognition approaches, focusing on umpire gestures allows the system to detect cricket events more reliably [1], [6], [11].

The detected events are then combined with scoreboard OCR outputs and player identification modules to generate structured event records with timestamps. These outputs enable the system to segment match videos into meaningful clips corresponding to important events such as boundaries and wickets.

D. Scoreboard OCR and Player Context Extraction

The proposed system uses EasyOCR to extract textual information from scoreboard overlays. This component plays a key role in identifying batsmen, bowlers, and score-related match context. The OCR outputs are cleaned and normalized to reduce noise caused by broadcast variations, text distortions, and overlay formatting.

```
Classification Report:
              precision    recall  f1-score   support

        four       0.90      0.84      0.87        32
   no_action       0.83      0.85      0.84        40
         out       0.83      0.78      0.81        51
         six       0.79      0.87      0.82        38

    accuracy                           0.83       161
   macro avg       0.84      0.84      0.84       161
weighted avg       0.83      0.83      0.83       161
```

Fig. 4. Classification report of Event Classification Model (ResNet-18)

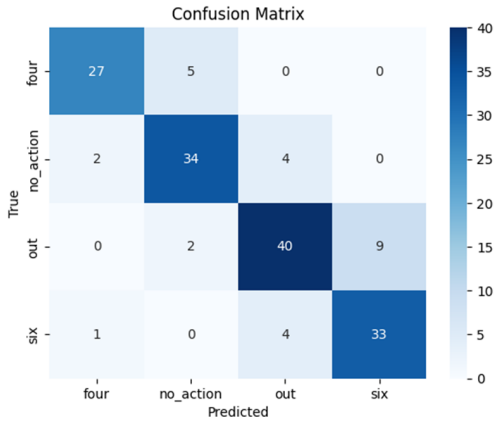

Fig. 5. Confusion matrix of Event Classification Model (ResNet-18)

The use of scoreboard text proved beneficial because it provided direct contextual information that is often difficult to infer from player appearance alone. Through OCR-based extraction, the system recovered information such as player names and score states, which were later associated with timestamps and detected events. This supports earlier findings that embedded textual content in sports broadcasts is highly useful for semantic video interpretation [7], [9].

In addition, the extracted OCR text was processed through a player identification pipeline that combines candidate generation, fuzzy matching against rosters, and Named Entity Recognition. This multimodal approach improved the reliability of associating detected events with the correct players, especially when OCR noise was present. The integration of textual and visual context demonstrates the practical value of multimodal sports video analysis [8], [12].

E. Event–Player Association and Video Segmentation

A key contribution of the proposed system is the synchronization of detected events and player information using timestamps. Once an event is detected from umpire signals, the system attaches the nearest player predictions extracted from scoreboard text. This enables the generation of structured event records containing the event type, event timestamp, batsmen, and bowler.

Additional post-processing modules were implemented to improve the usefulness of the final outputs. These include:

- ordinal event enrichment, which assigns sequence numbers such as first wicket or second four,
- wicket attribution, which estimates the dismissed batsman and wicket-taking bowler, and
- over segmentation, which identifies over boundaries based on bowler changes.

These enhancements make the generated cricket video segments more informative and easier to explore in downstream applications. Instead of simply tagging isolated events, the proposed system creates a richer structured representation of the match.

F. Semantic Indexing and Retrieval Performance

To support flexible retrieval, the extracted event records and transcript segments are transformed into sentence embeddings using MiniLM and indexed using FAISS for similarity search. This allows the system to support natural language retrieval over indexed cricket video content. For example, users can search for player appearances, wickets by a specific bowler, or ordinal events such as the second wicket or the last six.

The use of sentence embeddings for semantic retrieval is consistent with prior work on Sentence-BERT, which demonstrates the effectiveness of embedding-based similarity for semantic matching [4]. Likewise, FAISS provides an efficient and scalable indexing mechanism for approximate nearest-neighbor retrieval over embedding spaces [3]. In the proposed system, these two components enable efficient retrieval of relevant match moments without requiring users to manually browse full-length videos.

The retrieval module therefore extends the system beyond event detection and transforms the extracted outputs into a searchable multimedia knowledge layer. This is particularly useful for analysts, coaches, and fans who need quick access to player-specific or event-specific segments.

G. Web-Based Platform and Practical Usability

The final outputs of the system were integrated into a web-based platform developed using a backend–frontend architecture. The platform supports:

- player-wise filtering
- event-based navigation
- over-based browsing
- transcript-assisted exploration
- wicket attribution display
- semantic search with direct video playback links

The platform allows users to upload a cricket match video, process it through the backend pipeline, and interactively explore the generated outputs. The interface provides timestamped video navigation and supports direct jumping to relevant moments. This makes the system more practical than a standalone offline model because users can benefit from the results through an accessible application layer.

The inclusion of semantic search and structured indexing within an interactive platform addresses a limitation observed in many previous sports video analysis systems, where detection and retrieval are often treated as separate tasks. By combining OCR-based context extraction, event recognition, semantic indexing, and web-based interaction, the proposed

framework offers a more complete end-to-end solution for cricket video understanding [8].

H. Discussion

The experimental results show that the proposed system is effective as a multimodal framework for cricket video segmentation. The umpire detection model achieved strong test performance, validating the use of EfficientNet-B0 as a filtering stage before event recognition. The pose classifier based on ResNet-18 further demonstrated the feasibility of detecting cricket events directly from umpire gestures. At the same time, OCR-based scoreboard extraction provided important contextual support for identifying players and associating them with events.

A major strength of the proposed work is that it does not rely on a single modality. Instead, it combines visual event recognition, OCR-based textual context, NER-based player identification, timestamp synchronization, and semantic retrieval. This multimodal design improves robustness and reflects the advantages reported in recent multimedia and sports video analysis research [1], [7], [8], [12].

Another important outcome is the practical usability of the final system. The proposed architecture not only detects events but also transforms them into searchable, timestamped segments through MiniLM embeddings and FAISS indexing [3], [4]. This improves accessibility for end users and reduces the manual effort required to locate meaningful moments in long cricket recordings.

However, some limitations remain. OCR quality may degrade under poor broadcast graphics, motion blur, or low-resolution overlays. Similarly, pose-based event classification can be affected by occlusions, non-standard camera angles, or visually ambiguous gestures. Future improvements may include stronger OCR refinement, temporal modeling across consecutive frames, larger datasets for umpire pose recognition, and more advanced entity linking strategies for player identification.

Overall, the results demonstrate that the proposed system provides a practical and technically sound approach for automated cricket video segmentation and retrieval, while also offering a foundation for future enhancements in multimodal sports video intelligence.

V. Conclusion

This research presented an automated cricket video segmentation system for identifying player-related events in broadcast cricket videos. The proposed approach combines scoreboard text extraction using EasyOCR, player identification using NER-based processing, and umpire-signal-based event detection using EfficientNet-B0 and ResNet-18. The detected events and player information are synchronized through timestamps to generate structured video segments. In addition, semantic retrieval using MiniLM embeddings and FAISS enables efficient search and navigation through processed match content. The results show that the proposed multimodal approach can reduce manual effort in cricket video analysis while improving access to important player-specific and event-based moments. Future work can focus on improving OCR robustness, increasing pose dataset diversity, and enhancing temporal modeling for more accurate event detection.

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